Open Orthophosphate Model

River Deep Mountain AI

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Abstract

Orthophosphate (OrthoP) is a primary indicator of nutrient pollution in freshwater ecosystems, fundamentally impacting eutrophication and biodiversity. Current monitoring frameworks are constrained by high costs and spatiotemporal sparsity, as OrthoP measurements constitute approximately 2% of the Environment Agency's (EA) water quality records [1].

This paper details the Open Orthophosphate Model, an artificial intelligence-based framework utilising 24 years of EA monitoring data comprising 70 million observations [1].

The methodology integrates a hybrid feature selection process, including chemical clustering via Morgan Fingerprints and spatiotemporal encoding. The resulting LightGBM model utilises 45 representative features to estimate OrthoP concentrations across English river catchments. Performance evaluations indicate a Normalised Root Mean Square Error (NRMSE) of approximately 8%, demonstrating acceptable predictive accuracy. Furthermore, the framework incorporates SHapley Additive exPlanations (SHAP) analysis to ensure transparency and compliance with Responsible AI principles.

1 Introduction

The Open Orthophosphate Model addresses a monitoring gap in the United Kingdom where orthophosphate measurements represent only 2% of national records. Utilising 45 surrogate determinands identified through chemical clustering and domain expertise, the model predicts concentrations across areas with no orthophosphate monitoring.

Key gaps addressed:

- **Observation Sparsity:** OrthoP and phosphorus account for only 2% and 0.2% of national records, respectively
- **Monitoring Scale:** The model enables prediction at 72% of monitoring points at which OrthoP is never or rarely monitored.

This open-source framework provides spatial insights into nutrient levels, supporting regulatory compliance through a transparent, reproducible methodology.

1.1 River Deep Mountain AI

River Deep Mountain AI is an innovation project funded by the Ofwat Innovation Fund working collaboratively to develop open-source AI/ML models that can inform effective actions to tackle waterbody pollution.
The project consists of 6 core partners: Northumbrian Water, Cognizant Ocean, Xylem Inc., Water Research Centre Limited, The Rivers Trust and ADAS. The project is further supported by 6 water companies across the United Kingdom and Ireland.

2 Background and Context

Orthophosphate concentrations are critical to aquatic health, as excessive levels contribute to eutrophication and biodiversity loss. As directed by the Water Framework Directive [2] (WFD), the Water Environment Regulations (England and Wales) mandate monitoring for the River Basing Management Plan (RBMP) cycle and compliance. However, traditional sampling remains geographically and temporally limited and resource intensive. Machine learning offers a scalable alternative by inferring orthophosphate levels from correlated physio-chemical parameters and spatiotemporal features, leveraging historical data to bridge existing monitoring gaps.

3 Objective

The primary objective of this model is the estimation of orthophosphate concentrations using cost-effective, readily available sensory and spatiotemporal metrics. This addresses the challenge of monitoring nutrient levels across extensive catchments where historical data remains sparse. Such capability facilitates proactive environmental management and the optimisation of monitoring resources.

4 Data Architecture and Pre-processing

The integrity of the Open Orthophosphate Model relies on a high-volume, multi-source data architecture. This section outlines the EA Water Quality Archive dataset utilised [1], the methods for data harmonisation, and the rigorous data cleaning protocols applied to ensure model robustness.

4.1 Data Sources

The primary dataset is the Environment Agency (EA) Water Quality Archive, spanning 24 years (2000–2024) [1]. It contains approximately 70 million observations across 64,000 sampling points and 3,261 chemical determinands.

Step	Records removed	Percentage
Outlier Filtering	1.2M	1.7%
Duplicate Removal	0.8M	1.1%
Post-Imputation Size	68M	–

Table 1: Data quality and cleaning summary

The distribution of orthophosphate measurements are detailed in Fig.1. While surface water samples from rivers form the core training set, supplementary data from citizen science initiatives and high-frequency EA sonde sensors were utilised for validation (see section 8).

4.2 Data Quality and Cleaning

Stringent QA/QC protocols were implemented to minimise noise and handle the non-linear nature of the dataset.

- **Outlier Detection:** Applying Interquartile Range (IQR) methods and domain-specific cap. This included the implementation of a 0.5 mg/l cap for orthophosphate readings to exclude anomalous data. The limitations of this cap are highlighted in section 8.
- **Filtered Focus:** Reducing the sample size to 33.42 million by focusing on surface water and removing low-frequency determinands.
- **Missing values**: K-Nearest Neighbor (KNN) was applied to infill missing values in surrogate determinands.

Additionally, non-influential records, such as beach signage or irrelevant metadata, were removed to reduce dataset noise.

5 Model development

The development phase focused on transforming raw water quality data into predictive features using a combination of chemical informatics and statistical modeling. This section details the pipeline from feature extraction to the final selection of the LightGBM architecture.

5.1 Feature Engineering

The transformation of raw monitoring data into predictive variables involved high-dimensional chemical analysis and spatiotemporal encoding. This process ensured that the model could interpret both the chemical properties of water constituents and the geographic context of the samples.

- **Molecular Fingerprinting**: 2,168 chemical determinands were converted into SMILES (Simplified Molecular Input Line Entry System)[3] notation and subsequently transformed into Morgan fingerprints[4]. These are bit-form representations (2,168 x 2,048 matrix) that capture the extended-connectivity fingerprints of chemical structures.
- **Dimensionality Reduction**: To manage high dimensionality, Principal Component Analysis (PCA) was applied to the fingerprints. The team selected 356 optimal components, retaining 93% variance to balance predictive power with the risk of overfitting.
- **Spatiotemporal Encoding**:
  - **Spatial**: Features were derived from British National Grid (BNG) coordinates and geohash representations to enable hierarchical spatial indexing of the 64,000 sampling points.
  - **Temporal**: The model incorporates Season, Day of Week, and Month of Year to account for cyclical nutrient dynamics.
- **Environmental Context**: Supplemental features such as water temperature and river flow (derived from catchment GIS layers) were integrated to reflect ecological drivers of orthophosphate variability.

5.2 Feature Selection Strategy

The final selection of 45 representative features resulted from a hybrid methodology that combined unsupervised machine learning with statistical rigour. The full list of determinands and approach can be seen in our detailed model documentation available on GitHub[5].

- **Unsupervised Clustering**: Agglomerative hierarchical clustering was utilised to group determinands by molecular similarity, identifying two primary clusters directly related to phosphorus and orthophosphate.
- **Statistical Ranking**: Determinands were prioritised through:
  - **Spearman Ranking**: To identify non-linear associations.
  - **RFECV**: Recursive Feature Elimination with Cross-Validation was used to prune less impactful variables.
- **Mutual Information Regression**: This captured dependencies between features that standard correlation might miss.
- **Expert Curation**: Results were cross-verified by Subject Matter Experts (SMEs) to ensure the retention of ecologically significant determinands like nitrate, ammonia, pH, and dissolved oxygen.

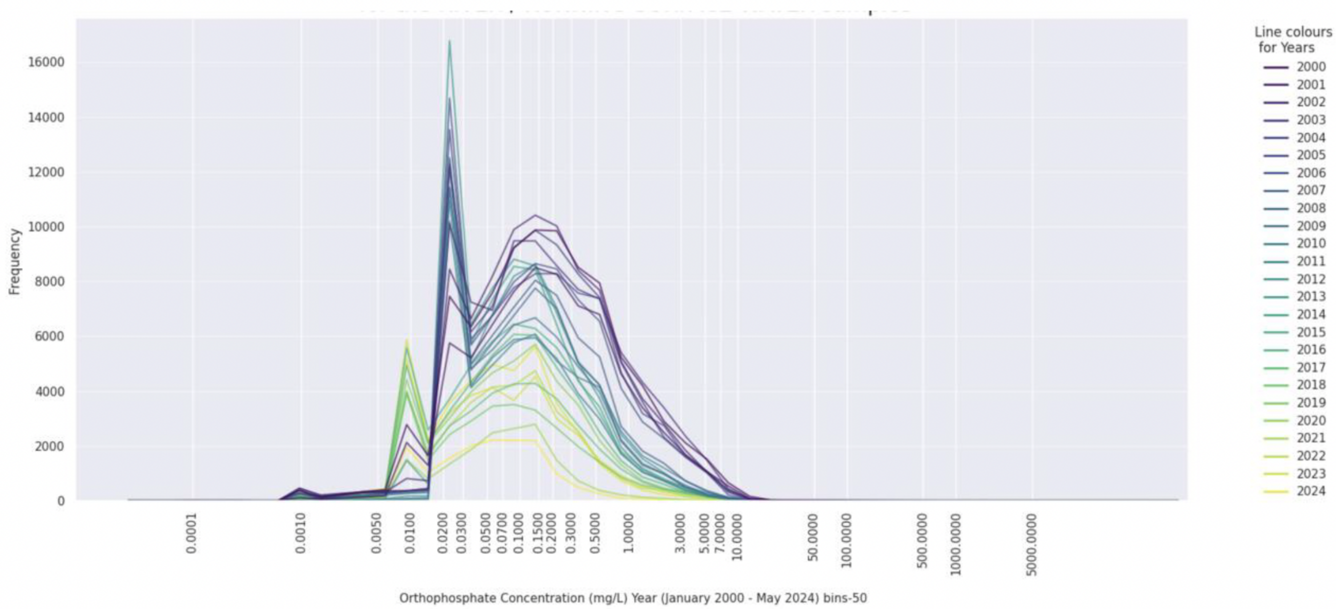

Figure 1: Distribution of orthophosphate samples from the EA Water Quality Archive. The observed frequency spikes are addressed in the limitations (Section 8).

5.3 Hyperparameter Optimisation

Optimal model performance was achieved through Bayesian optimisation. Parameters were tuned within specified ranges: a learning rate of 0.05–0.15, maximum tree depth of 8–12, and 64–128 leaves. To prevent overfitting, L1 and L2 regularisation penalties were applied, and early stopping was implemented after 100 rounds without validation loss improvement.

5.4 Explainability Framework (SHAP)

To ensure transparency, the SHAP (SHapley Additive exPlanations) framework was integrated[6]. This provides global interpretation by ranking feature importance across the dataset and local interpretation for specific sampling points. Visual outputs, such as beeswarm plots, offer stakeholders insight into the model's decision-making process.

5.5 Model Selection and Justification

Several algorithms, including XGBoost, Random Forest, and Artificial Neural Networks (ANN), were benchmarked:

- **Linear Regression**: Used as a baseline for performance.
- **Random Forest & XGBoost**: Evaluated for their ability to handle non-linear relationships.
- **Artificial Neural Networks (ANN)**: Tested for high-dimensional pattern recognition but found to pose higher risks of overfitting.
- **LightGBM:** Gradient boosting framework optimised for large datasets and sparse features.

LightGBM was selected as the final production model because it consistently provided the lowest residual errors (reaching a 7.89% NRMSE). The overview of model selection can be seen in our detailed model documentation available on GitHub[5].

LightGBM was preferred for its native handling of missing values and its superior performance on large, sparse tabular datasets compared to other gradient boosting or deep learning frameworks[7].

6 Model Performance

The performance of the model was established through a series of experiments that transitioned from broad national baseline assessments to high-resolution spatiotemporal catchments.

6.1 Key Metrics from Train and Test

As mentioned above, the LightGBM model was selected as the final architecture after benchmarking against four other models: XGBoost, Random Forest, Artificial Neural Networks (ANN), and Linear Regression.
LightGBM consistently yielded the lowest residual errors:

- Mean Squared Error (MSE) of 0.00088
- Mean Absolute Error (MAE) of 0.02
- Normalised Root Mean Square Error (NRMSE) of 7.89%

Catchment Type	RMSE	NRMSE	MSE	Percentage
All catchments	0.05	10.15%	0.002	0.03
High BFI	0.04	13.55%	0.002	0.03
Low BFI	0.07	14.98%	0.006	0.04
Natural	0.05	10.49%	0.003	0.03
Non-Natural	0.04	8.12%	0.001	0.03
Rural	0.08	16.59%	0.006	0.05
Urban	0.03	11.68%	0.001	0.02
Dry	0.06	13.09%	0.004	0.04
Wet	0.03	14.40%	0.001	0.02

Table 2: Overview of validation results using a full year of EA Water Quality Archive data from June 1, 2024 to July 17, 2025

The model achieved its primary objective by enabling orthophosphate predictions at 72% of unmonitored or rarely monitored locations, retrospectively. By training on historical data from 18,000+ sampling points, the model provides high-resolution insights for the remaining 44,000+ points across the national network.

6.2 Validation

The primary objective of the validation process was to evaluate the predictive accuracy and reliability of the LightGBM framework when exposed to datasets entirely independent of those used during training and testing. This process ensures the model's generalisability across diverse catchment types and real-world monitoring conditions.

6.2.1 Validation Using National Environment Agency Data

The model was validated using a full year of Environment Agency (EA) monitoring data from June 1, 2024, to July 17, 2025[1]. This "blind" test involved 216 sampling points across England that the model had not previously processed.

Approach: The model utilised its 45 finalised features, including Section 82 parameters and key determinands such as turbidity, conductivity, and nitrate-N.

Results: The validation confirmed that the model's performance remains consistent with its training metrics:

- Root Mean Square Error (RMSE): 0.05 mg/L
- MAE: 0.03 mg/L
- MSE: 0.00257 mg/L
- NRMSE: 102% (when calculated using the mean observed value), 10.15% (when calculated using the range of observations)

6.2.2 Performance Across Catchment Categories

To evaluate robustness across varied hydrological contexts, the model was tested against nine specific catchment categories.

Approach: Catchments were selected based on factors such as the Base Flow Index (BFI), land use (urban vs. rural), natural (mostly un-impacted by human influence) catchments, and precipitation levels (wet vs. dry).

Results: The model demonstrated moderate overall performance, with its highest accuracy observed in non-natural sites and urban catchments (see Table 2).

6.2.3 Out-of-Sample Validation: Citizen Science and Continuous Sondes

The framework was further tested against high-frequency and community-collected data to determine its utility as a decision-support tool.

Approach: Model predictions were compared against 81 orthophosphate measurements collected by citizen scientists at 6 sampling points, using proxy data from EA continuous sondes (e.g., ammonium as N) as model inputs[8].

Results: The validation revealed that the model performs more reliably at lower orthophosphate concentrations but struggles with high-concentration "spikes" (see Table 3).

This multi-faceted validation highlights the model's ability to generalise across standard monitoring scenarios while identifying specific areas, such as high-concentration rural catchments, where further refinement is needed.

7 Deployment Considerations

While the Open Orthophosphate Model provides a scalable framework for estimating nutrient concentrations, its

	Mean OrthoP (from Citizen Science)	**MAE** (mg/l)	$\mathbf{R^2}$	**RMSE**
All sampling points (81)	1.79 mg/l	1.72	0.025	2.32
Pix Brook (59)	2.3 mg/l	2.2	0.0007	2.69
River Severn (5)	0.078 mg/l	0.073	0.599	0.08
River Teme (3)	0.013 mg/l	0.052	1	0.05
Warren Dike (6)	0.166 mg/l	0.111	0.301	0.12
River Rede (1)	0.166 mg/l	0.0008	–	–
South Delph (7)	1.19 mg/l	1.054	0.054	1.06

Table 3: Results from Citizen Science and EA Sonde validation

application in regulatory and operational contexts requires a nuanced understanding of its design parameters and inherent limitations.

7.1 Advisory vs. Operational Utility

It is critical to distinguish between the intended use of the RDMAI programme models and their potential misapplication in high-stakes environments.

- **Advisory Status**: All models within the RDMAI programme are designed strictly for advisory purposes.
- **Regulatory Context**: While accurate measurement of orthophosphate is essential for regulatory compliance, this model should complement, rather than replace, traditional monitoring where high precision is legally required.
- **Integration**: The integration of this framework into existing monitoring platforms or decision-support systems is the sole responsibility of the end-user, whether a water utility or an advisory body.
- **Access**: To facilitate transparency and community-driven improvement, the entire source code and the trained model (stored as a pickle file) are available on GitHub[5]. Consequently, no centralized API endpoints have been developed.

8 Limitations and Uncertainty

The model's predictive reliability is heavily influenced by the quality and quantity of input data provided at the point of training and inference.

- **Input Sensitivity:** The model utilises 45 distinct determinands to generate predictions. While it can operate with missing data by defaulting to zero, the provision of at least 25 actual measured values is recommended to ensure robust reliability.
- **Spatial Dependency**: The model is optimised for use at existing Environment Agency (EA) sampling points. This constraints the model performance described above to England. If a prediction is requested for a location without an established WIMS (Water Quality Management System) ID, the model will run but will lack location-specific intelligence, likely degrading performance.
- **Concentration Caps**: The current iteration of the model was developed with a 0.5 mg/l cap on orthophosphate concentrations. Validation against citizen science data suggests the model struggles to accurately predict high-concentration "spikes" exceeding this threshold, and results in such contexts should be interpreted with caution.
- **Limit of Detection (LOD) Bias**: A significant portion (9%) of the EA monitoring records consists of "less than" values (e.g., $<$0.02 mg/l), representing the Limit of Detection (LOD). As these were included at face value during training, they might have introduced an artificial stratification of data in lower concentration ranges and thereby induced a potential bias into the model.

In addition, it is recommended that the model is re-trained routinely as new monitoring data becomes available.

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Disclaimer

This document has been prepared by the RDMAI consortium and is intended for informational purposes only.

The model described in this document is the product of an ongoing research and development project. It represents an early iteration and its outputs are preliminary. The model is not intended for use in production environments, regulatory reporting, environmental monitoring, or any decision-making in relation to public health, public policy, or commercial matters without independent validation and the acceptance of full liability by the user.

All use of the model and its associated software is subject to the full terms set out in the RDMAI repository, available at https://github.com/Cognizant-RDMAI.

Readers seeking technical detail on model performance, limitations, and data sources should refer to the accompanying Model Report. The RDMAI consortium and its members accept no liability for any outcomes arising from use of or reliance on the model or its outputs.